Federated learning

Unit 9 · reference sheet · AI/ML in 35 topics

Protocol

1. broadcast w_t 2. local SGD $\times E$ 3. upload w_t^k 4. aggregate

$w_{t+1} = \sum_k \frac{n_k}{n} w_t^k \qquad n = \sum_k n_k$

Communication is the bottleneck: rounds × clients × model size.

Non-IID

Client drift: local optima diverge; the average fits nobody.

FedProx: $F_k(w) + \frac{\mu}{2}\|w - w_t\|^2$ (larger $\mu \Rightarrow$ stronger leash)

$E \uparrow \Rightarrow$ fewer rounds, more drift. Not monotone.

Report per-client accuracy; a high mean can hide a small client.

Privacy

DP: $P(M(D) \in S) \leq e^{\varepsilon} P(M(D') \in S) + \delta$

DP-SGD: clip per-example gradient to C, add $\mathcal{N}(0, \sigma^2 C^2)$

Budget composes over rounds ($10 \times 0.1 \approx 1.0$ basic).

Secure aggregation: server sees only the sum; no protection from malicious clients.

Threats

membership inference, gradient inversion, poisoning, backdoor

Defences: clipping + noise, robust aggregation (median/trimmed), update inspection

Gotchas

- Federated ≠ private. Gradients are a fingerprint.
- FedAvg ≠ pooled training unless clients are IID.
- More local epochs is not free: drift grows with E.

This unit at a glance

#	Topic	You can
33	Federated setting & FedAvg	describe the round protocol, write the averaging rule, justify data locality
34	Non-IID data & convergence	explain client drift and name the fixes; state the local-steps trade-off
35	Privacy & threat model	explain DP noise, secure aggregation, and poisoning/backdoor attacks

If you see this → do that

Signal	Move
Data cannot move	federated
Clients differ	FedProx / personalise
Privacy claim	DP + secure agg
Malicious clients	robust aggregation
Too many rounds	more local epochs (drift!)

Numbers worth memorising

FedAvg weights 100/200/700 $\Rightarrow$ 0.1/0.2/0.7

Weighted mean of $0.1, 0.5, 0.9 \rightarrow 0.74$ (equal: 0.50)

DP: clip C; noise σC; 10×0.1 budget ≈ 1.0

E local epochs: rounds ↓, drift ↑